Linear programming

- Example Numpy: PageRank
- Linear programming (LP): scipy.optimize.linprog
- Example linear programming: Maximum flow
- Mixed integer linear programming (MILP)

PageRank

- Google's original search engine ranked webpages using PageRank

The Anatomy of a Large-Scale Hypertextual Web Search Engine

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Abstract

In this paper, we present Google, a prototype of a large-scale search engine which makes heavy use of the structure present in hypertext. Google is designed to crawl and index the Web efficiently and produce much more satisfying search results than existing systems. The prototype with a full text and hyperlink database of at least 24 million pages is available at http://google.stanford.edu/ To engineer a search engine is a challenging task. Search engines index tens to hundreds of millions of web pages involving a comparable number of distinct terms. They answer tens of millions of queries every day. Despite the importance of large-scale search engines on the web, very little academic research has been done on them. Furthermore, due to rapid advance in technology and web proliferation, creating a web search engine today is very different from three years ago. This paper provides an in-depth description of our large-scale web search engine -- the first such detailed public description we know of to date. Apart from the problems of scaling traditional search techniques to data of this magnitude, there are new technical challenges involved with using the additional information present in hypertext to produce better search results. This paper addresses this question of how to build a practical large-scale system which can exploit the additional information present in hypertext. Also we look at the problem of how to effectively deal with uncontrolled hypertext collections where anyone can publish anything they want.

Keywords

World Wide Web, Search Engines, Information Retrieval, PageRank, Google

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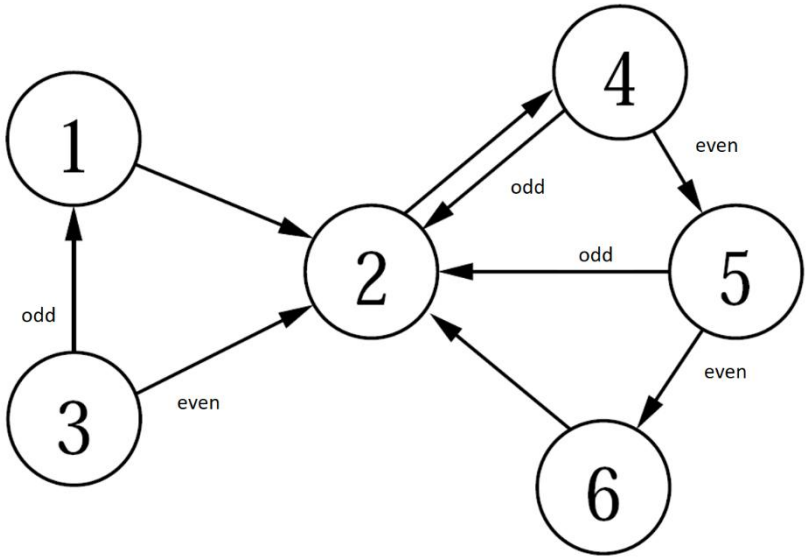
DOI 10.1016/S0169-7552(98)00110-X / infolab.stanford.edu/pub/papers/google.pdf

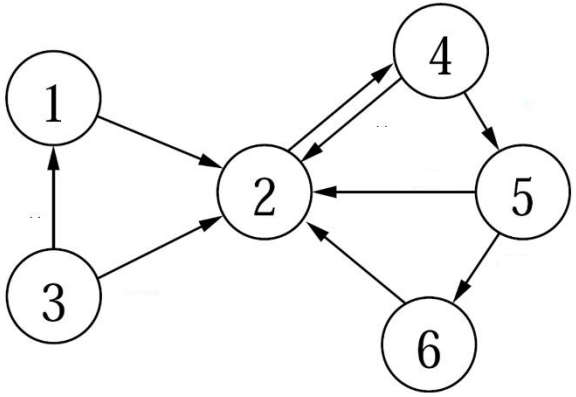
PageRank - A NumPy / Jupyter / matplotlib example

- View the internet as a graph where nodes correspond to webpages and directed edges to links from one webpage to another webpage



Five different ways to compute PageRank probabilities

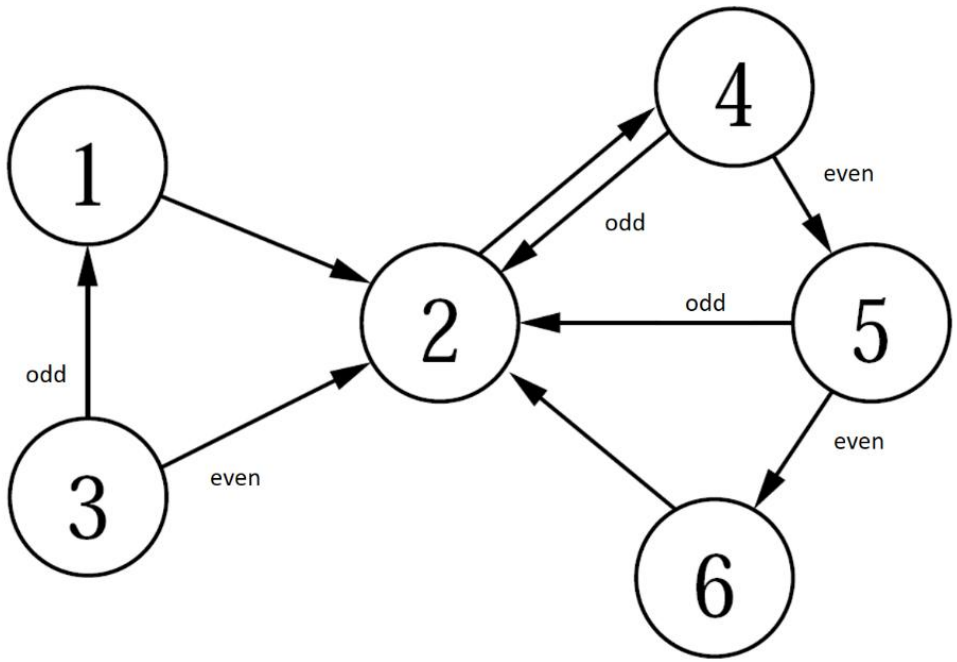
1) Simulate random process manually by rolling dices
2) Simulate random process in Python
3) Computing probabilities using matrix multiplication
4) Repeated matrix squaring
5) Eigenvector for $\lambda = 1$



Random surfer model (simplified)

The PageRank of a node (web page) is the fraction of the time one visits a node by performing an *infinite random traversal* of the graph starting at node 1, and in each step

- with probability 1/6 jumps to a random page (probability 1/6 for each node)
- with probability 5/6 follows an outgoing edge to an adjacent node (selected uniformly)

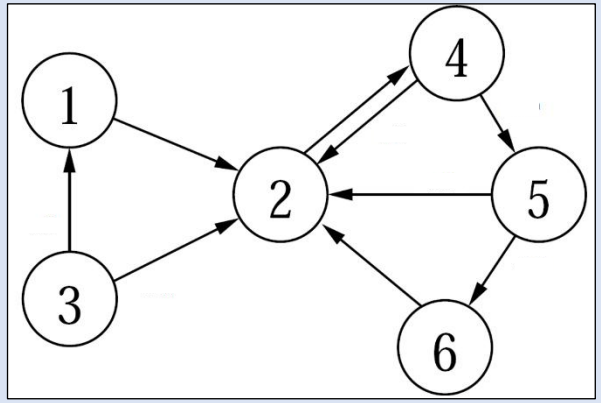


The above can be simulated by using a dice: Roll a *dice*. If it shows 6, jump to a random page by rolling the dice again to figure out which node to jump to. If the dice shows 1-5, follow an outgoing edge - if two outgoing edges roll the dice again and go to the lower number neighbor if it is odd.

Adjacency matrix and degree vector

pagerank.ipynb

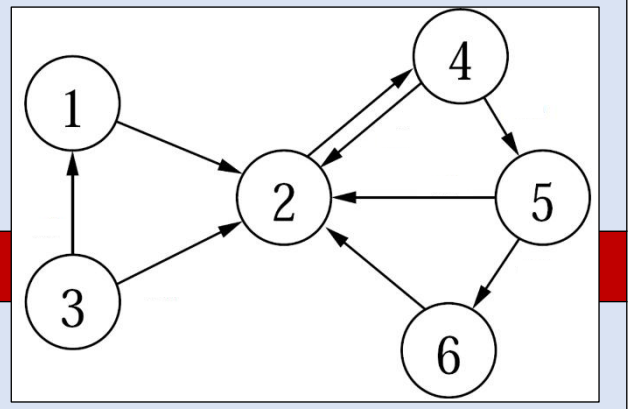
```
import numpy as np

# Adjacency matrix of the directed graph in the figure
# (note that the rows/columns are 0-indexed, whereas in the figure the nodes are 1-indexed)

              # 1  2  3  4  5  6
G = np.array([[0, 1, 0, 0, 0, 0],  # 1
              [0, 0, 0, 1, 0, 0],  # 2
              [1, 1, 0, 0, 0, 0],  # 3
              [0, 1, 0, 0, 1, 0],  # 4
              [0, 1, 0, 0, 0, 1],  # 5
              [0, 1, 0, 0, 0, 0]]) # 6

n = G.shape[0]  # number of rows in G
degree = np.sum(G, axis=1, keepdims=True)  # column vector with row sums = out-degrees

# The below code handles sinks, i.e. nodes with out-degree zero (no effect on the graph above)

G = G + (degree == 0)  # add edges from sinks to all nodes (uses broadcasting)
degree = np.sum(G, axis=1, keepdims=True)
```



Simulate random walk (random surfer model)

pagerank.ipynb

```
from random import randint, choice

STEPS = 1000000

# adjacency_list[i] is a list of all j where (i, j) is an edge of the graph.
adjacency_list = [[j for j, e in enumerate(row) if e] for row in G]

count = np.zeros(n)          # histogram over number of node visits
state = 0                    # start at node with index 0
for _ in range(STEPS):
    count[state] += 1        # increment count for state
    if randint(1, 6) == 6:   # original paper uses 15% instead of 1/6
        state = randint(0, 5)
    else:
        state = choice(adjacency_list[state])

print(f'{adjacency_list = }', count / STEPS, sep='\n')
```



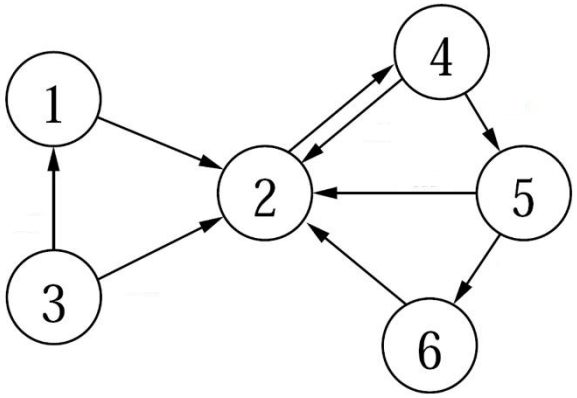
Python shell

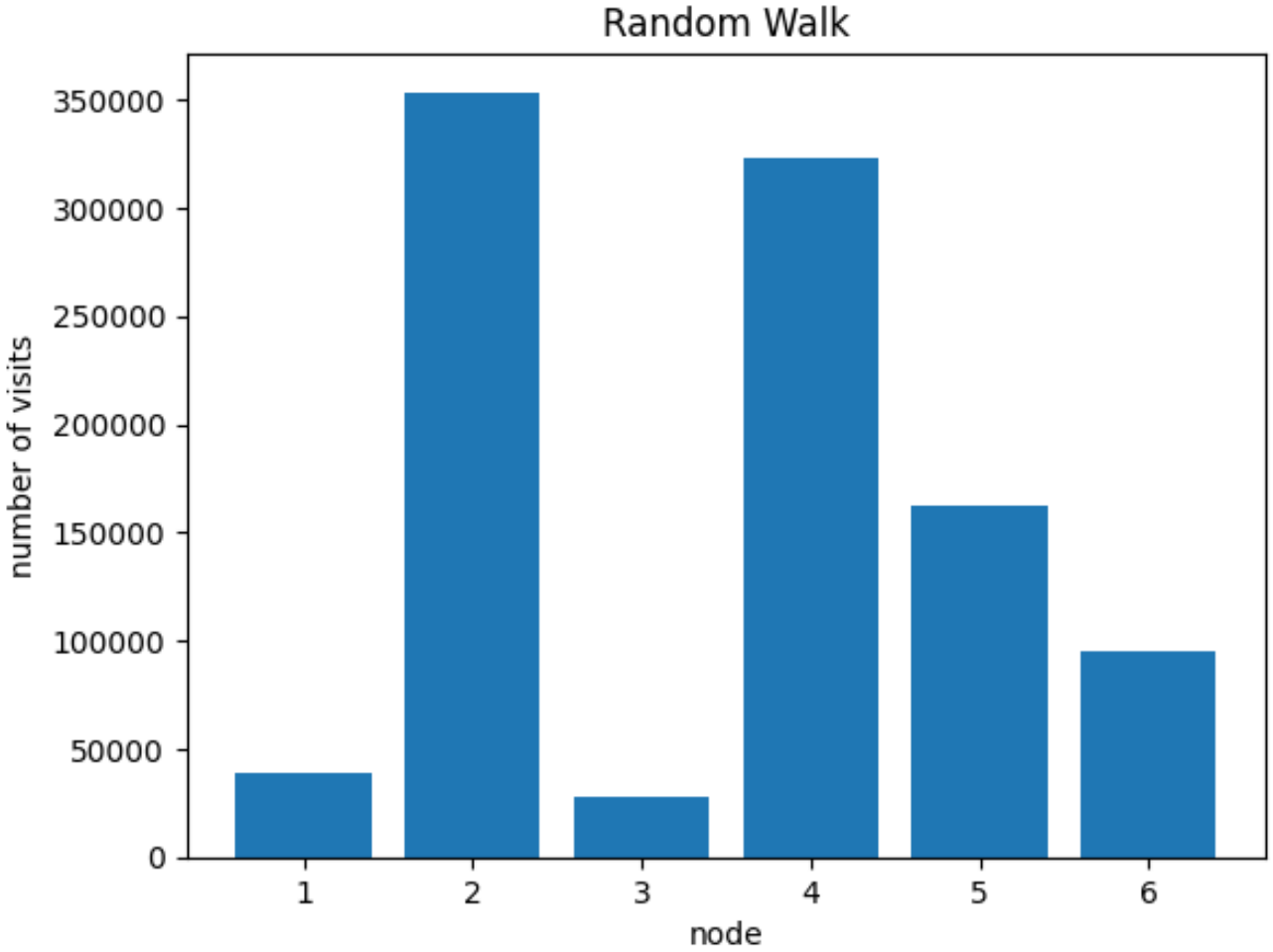
```
| adjacency_list = [[1], [3], [0, 1], [1, 4], [1, 5], [1]]
| [0.039365 0.353211 0.02751  0.322593 0.1623   0.095021]
```

Simulate random walk (random surfer model)

pagerank.ipynb

```
import matplotlib.pyplot as plt
plt.bar(range(1, 7), count)
plt.title('Random Walk')
plt.xlabel('node')
plt.ylabel('number of visits')
plt.show()
```





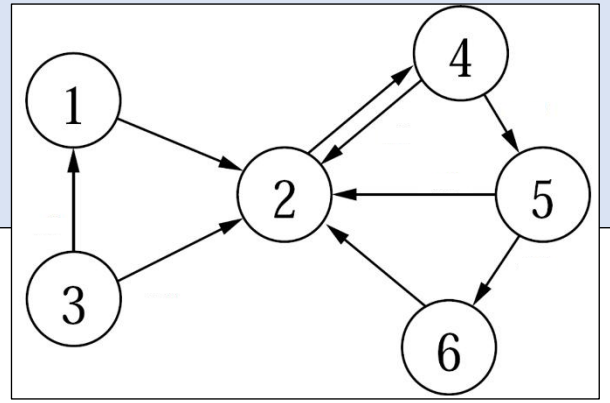
Transition matrix *A*

pagerank.ipynb

```
A = G / degree   # Normalize row sums to one. Note that 'degree'
                 # is an n x 1 matrix, whereas G is an n x n matrix.
                 # The elementwise division is repeated for each column of G
print(A)
```

Python shell

```
[[0.   1.   0.   0.   0.   0. ]
 [0.   0.   0.   1.   0.   0. ]
 [0.5  0.5  0.   0.   0.   0. ]
 [0.   0.5  0.   0.   0.5  0. ]
 [0.   0.5  0.   0.   0.   0.5]
 [0.   1.   0.   0.   0.   0. ]]
```



Repeated matrix multiplication

We now want to compute the probability $p^{(i)}_j$ to be in vertex j after i steps. Let $p^{(i)} = (p^{(i)}_0, \dots, p^{(i)}_{n-1})$.

Initially we have $p^{(0)} = (1, 0, \dots, 0)$.

We compute a matrix M, such that $p^{(i)} = M^i \cdot p^{(0)}$ (assuming $p^{(0)}$ is a column vector).

If we let $\mathbf{1}_n$ denote the $n \times n$ matrix with 1 in each entry, then M can be computed as:

$$p_j^{(i+1)} = \frac{1}{6} \cdot \frac{1}{n} + \frac{5}{6} \sum_k p_k^{(i)} \cdot A_{k,j}$$

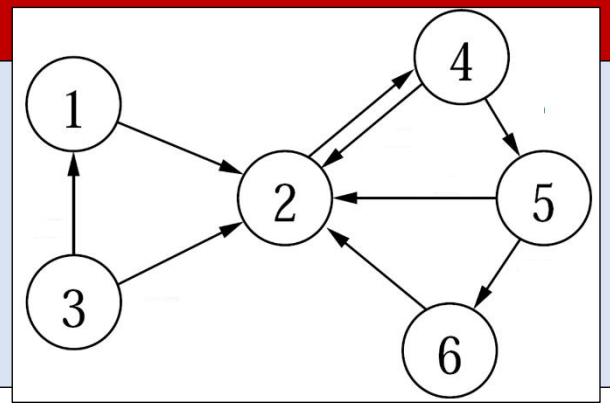
$$p^{(i+1)} = \underbrace{\left(\frac{1}{6} \cdot \frac{1}{n} \mathbf{1}_n + \frac{5}{6} A^{\mathsf{T}} \right)}_{M} \cdot p^{(i)}$$

pagerank.ipynb

```
ITERATIONS = 20
p_0 = np.zeros((n, 1))
p_0[0, 0] = 1.0
M = 1 / (6 * n) + 5 / 6 * A.T
p = p_0
prob = p   # 'prob' will contain each
           # computed 'p' as a new column
for _ in range(ITERATIONS):
    p = M @ p
    prob = np.append(prob, p, axis=1)
print(p)
```

Python shell

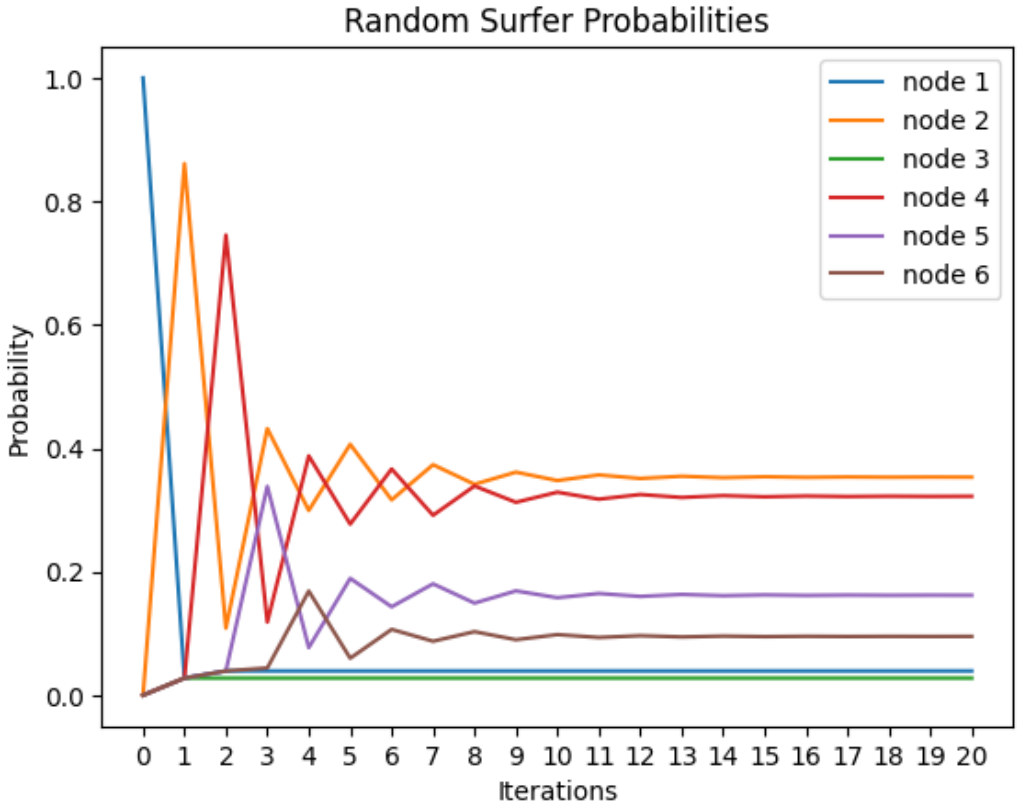
```
[[0.03935185]
 [0.35326184]
 [0.02777778]
 [0.32230071]
 [0.16198059]
 [0.09532722]]
```

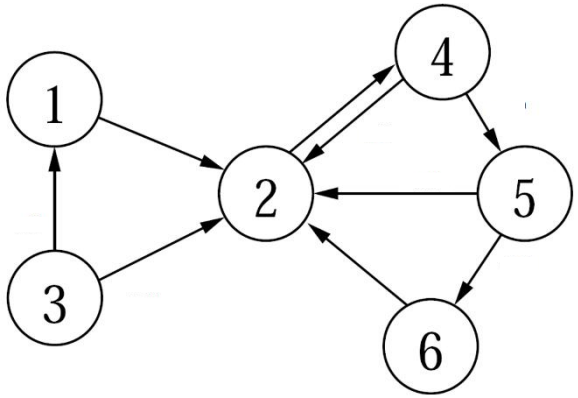


Rate of convergence

pagerank.ipynb

```
x = range(ITERATIONS + 1)
for node in range(n):
    plt.plot(x, prob[node], label=f'node {node + 1}')
plt.xticks(x)
plt.title('Random Surfer Probabilities')
plt.xlabel('Iterations')
plt.ylabel('Probability')
plt.legend()
plt.show()
```





Repeated squaring

$$\underbrace{M\cdot(\cdots(M\cdot(M}_{k \text{ multiplications, } k \text{ power of 2}}\cdot p^{(0)}))\cdots) = M^k\cdot p^{(0)} = M^{2^{\log_2 k}}\cdot p^{(0)} = (\cdots((\overbrace{M^2)^2)^2\cdots)^2}^{\log_2 k}\cdot p^{(0)}$$

pagerank.ipynb

```
from math import log2
MP = M
for _ in range(1 + int(log2(ITERATIONS))):
    MP = MP @ MP
p = MP @ p_0
print(p)
```

Python shell

```
[[0.03935185]
 [0.35332637]
 [0.02777778]
 [0.32221711]
 [0.16203446]
 [0.09529243]]
```

PageRank : Computing eigenvector for λ = 1

- We want to find a vector p, with $|p| = 1$, where $M \cdot p = p$, i.e. an *eigenvector* p for the eigenvalue $\lambda = 1$

pagerank.ipynb

```
eigenvalues, eigenvectors = np.linalg.eig(M)

idx = eigenvalues.argmax()              # find the largest eigenvalue (= 1)
p =  np.real(eigenvectors[:, idx])      # .real returns the real part of complex numbers
p /= p.sum()                            # normalize p to have sum 1

print(p)
```

Python shell

```
| [0.03935185 0.3533267  0.02777778 0.32221669 0.16203473 0.09529225]
```

PageRank : Note on practicality

- In practice an explicit $n \times n$ matrix for billions of nodes is infeasible, since the number of entries would be order of 10^{18}

- Instead use sparse matrices (in Python modul `scipy.sparse`) and stay with repeated multiplication (since fast convergence)

Linear programming (LP): scipy.optimize.linprog

scipy.optimize.linprog

- scipy.optimize.linprog can solve *linear programs* of the following form, where one wants to find an n x 1 vector x satisfying:

Minimize: $c^T \cdot x$

Subject to: $A_{ub} \cdot x \leq b_{ub}$
$A_{eq} \cdot x = b_{eq}$

dimension

$c : n \times 1$

$A_{ub} : m \times n$ $\quad b_{ub} : m \times 1$

$A_{eq} : k \times n$ $\quad b_{eq} : k \times 1$

Some other open-source optimization libraries PuLP and Pyomo

For industrial strength linear solvers, use solvers like Cplex or Gurobi (mixed-integer linear programs)

Linear programming (LP) example

Maximize

$3 \cdot x_1 + 2 \cdot x_2$

Subject to

$2 \cdot x_1 + 1 \cdot x_2 \leq 10$

$5 \cdot x_1 + 6 \cdot x_2 \geq 4$

$-3 \cdot x_1 + 7 \cdot x_2 = 8$

$\Updownarrow$

Minimize

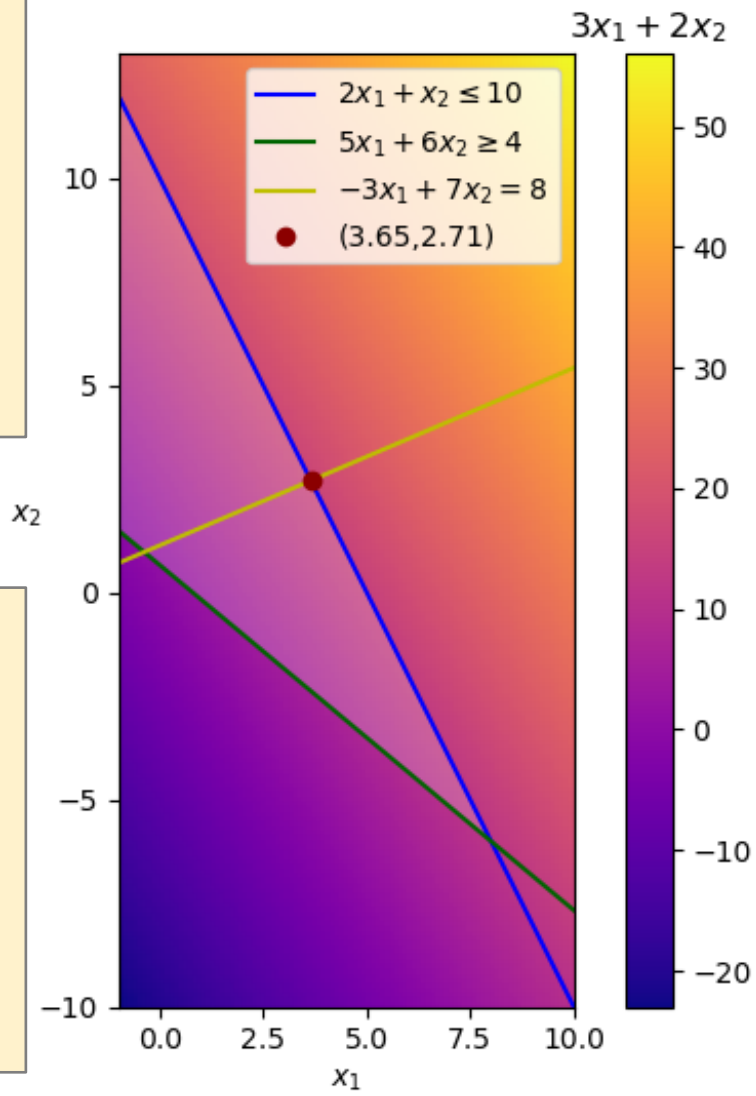
$-(3 \cdot x_1 + 2 \cdot x_2)$

Subject to

$2 \cdot x_1 + 1 \cdot x_2 \leq 10$

$-5 \cdot x_1 + -6 \cdot x_2 \leq -4$

$-3 \cdot x_1 + 7 \cdot x_2 = 8$



linear_programming.py

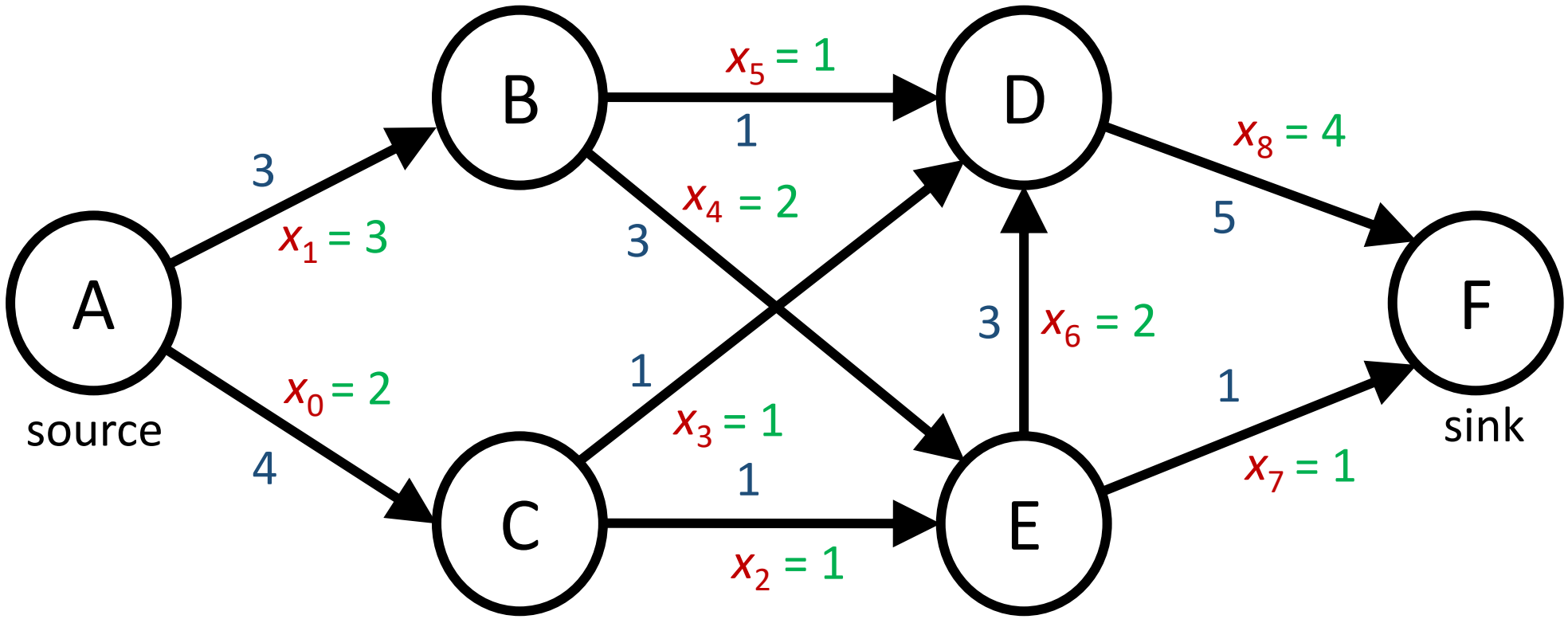
```
import numpy as np
from scipy.optimize import linprog
c = np.array([3, 2])
A_ub = np.array([[ 2,  1],
                 [-5, -6]])  # multiplied by -1
b_ub = np.array([10, -4])
A_eq = np.array([[-3, 7]])
b_eq = np.array([8])
res = linprog(-c,  # maximize = minimize the negated
              A_ub=A_ub,
              b_ub=b_ub,
              A_eq=A_eq,
              b_eq=b_eq)
print(res)  # res.x is the optimal vector
```

Python shell

```
  message: Optimization terminated successfully.
           (HiGHS Status 7: Optimal)
  success: True
   status: 0
      fun: -16.352941176470587
        x: [ 3.647e+00  2.706e+00]
  ...
```

Example linear programming (LP): Maxmium flow

Solving maximum flow using linear programming



Maximize

$x_7 + x_8$ — flow value

Subject to

capacity constraints:

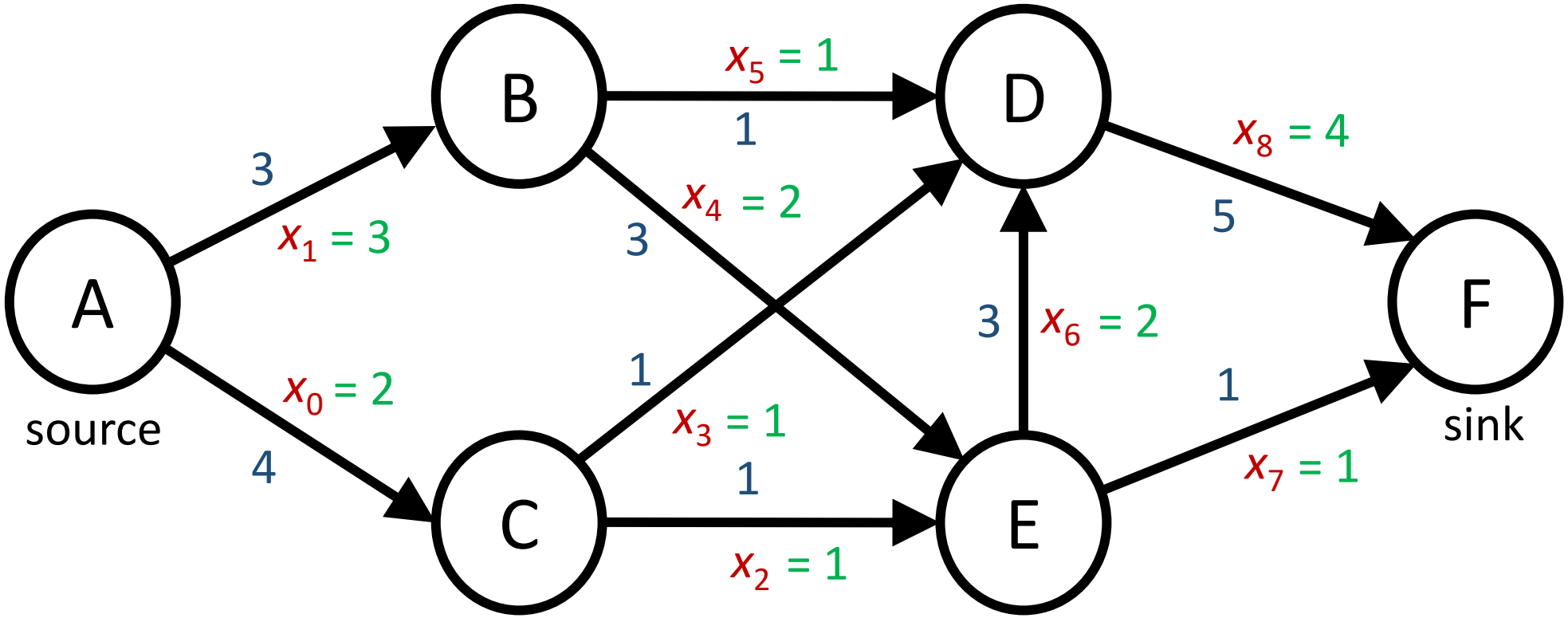
$x_0 \le 4$
$x_1 \le 3$
$x_2 \le 1$
$x_3 \le 1$
$x_4 \le 3$
$x_5 \le 1$
$x_6 \le 3$
$x_7 \le 1$
$x_8 \le 5$

flow conservation:

$x_1 = x_4 + x_5$
$x_0 = x_2 + x_3$
$x_3 + x_5 + x_6 = x_8$
$x_2 + x_4 = x_6 + x_7$

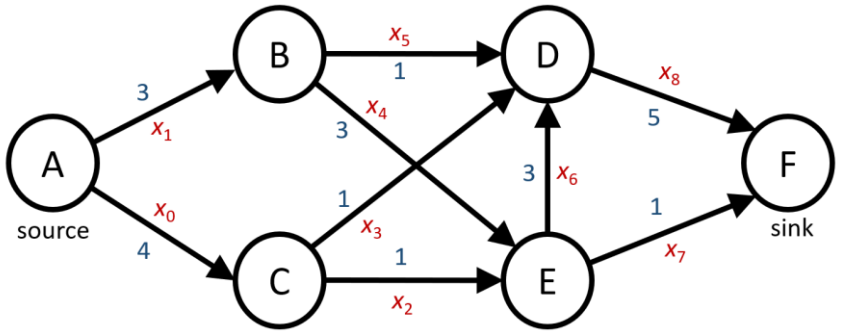
We will use the scipy.optimize.linprog function to solve the *maximum flow* problem on the above directed graph. We want to send as much *flow* from node A to node F. Edges are numbered 0..8 and each edge has a maximum *capacity*.

Note: solution not unique


A
B
C
D
E
F
source
sink
3
$x_1 = 3$
$x_0 = 2$
4
$x_5 = 1$
1
$x_4 = 2$
3
1
$x_3 = 1$
1
$x_2 = 1$
3
$x_6 = 2$
$x_8 = 4$
5
1
$x_7 = 1$

Solving maximum flow using linear programming

- $x = (x_0,\ldots,x_8)$ is a vector describing the flow along each edge
- c is a vector to add the flow along the edges (7 and 8) to the sink (F), i.e. a function computing *the flow value*
- A_{ub} and b_{ub} is a set of *capacity constraints*, for each edge flow ≤ capacity
- A_{eq} and b_{eq} is a set of *flow conservation* constraints, for each non-source and non-sink node (B, C, D, E), requiring that the flow into equals the flow out of a node



Minimize

$\boxed{-x_7 - x_8}$ $c^T \cdot x$ flow value

Subject to

$x_0 \le 4$
$x_1 \le 3$
$x_2 \le 1$
$x_3 \le 1$
$x_4 \le 3$
$x_5 \le 1$
$x_6 \le 3$
$x_7 \le 1$
$x_8 \le 5$

$A_{ub} \cdot x \le b_{ub}$
$\Updownarrow$
$I \cdot x \le$ capacity

capacity constraints

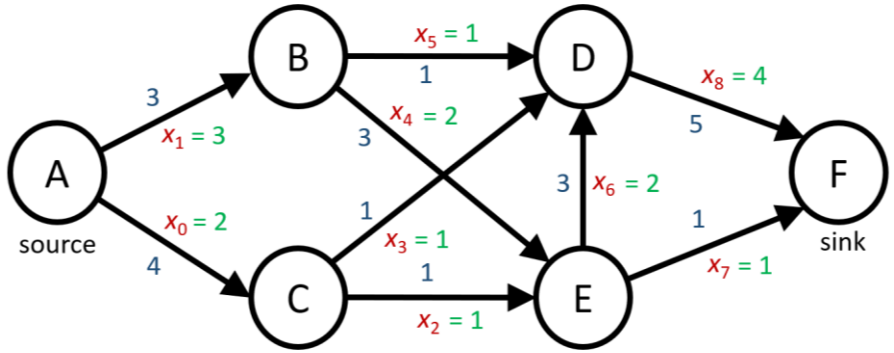
$A_{eq} \cdot x = b_{eq} = 0$

$0 = -x_1 + x_4 + x_5$
$0 = -x_0 + x_2 + x_3$
$0 = -x_3 - x_5 - x_6 + x_8$
$0 = -x_2 - x_4 + x_6 + x_7$

flow conservation

maximum-flow.py

```python
import numpy as np
from scipy.optimize import linprog

#                                0  1  2  3  4  5  6  7  8
conservation = np.array([[ 0,-1, 0, 0, 1, 1, 0, 0, 0],  # B
                         [-1, 0, 1, 1, 0, 0, 0, 0, 0],  # C
                         [ 0, 0, 0,-1, 0,-1,-1, 0, 1],  # D
                         [ 0, 0,-1, 0,-1, 0, 1, 1, 0]]) # E

#                    0  1  2  3  4  5  6  7  8
sinks = np.array([0, 0, 0, 0, 0, 0, 0, 1, 1])

#                      0  1  2  3  4  5  6  7  8
capacity = np.array([4, 3, 1, 1, 3, 1, 3, 1, 5])

res = linprog(-sinks,
              A_eq=conservation,
              b_eq=np.zeros(conservation.shape[0]),
              A_ub=np.eye(capacity.size),
              b_ub=capacity)

print(res)
```



Python shell

```
  message: Optimization terminated successfully.
           (HiGHS Status 7: Optimal)
  success: True
   status: 0
      fun: -5.0
        x: [ 2.000e+00  3.000e+00  1.000e+00  1.000e+00  2.000e+00
             1.000e+00  2.000e+00  1.000e+00  4.000e+00]
```

the solution found varies with the scipy version

Mixed integer linear programming (MILP)

Mixed integer linear programming (MILP)

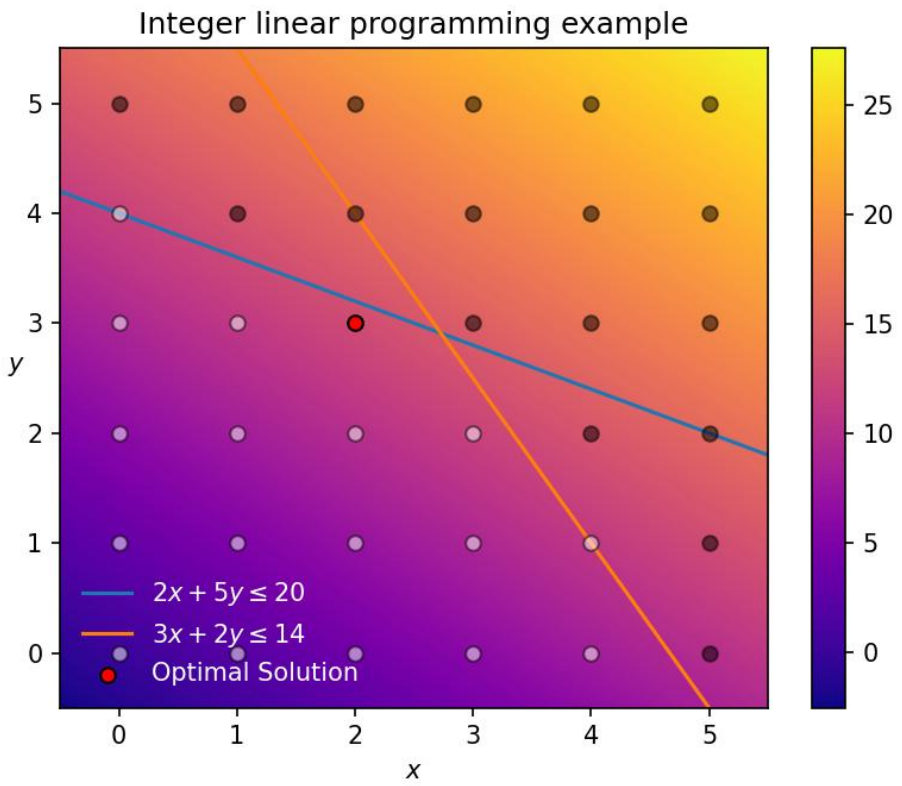
- Combines arithmetic with logic (binary, integer and continuous variables)
- Open source solver: scipy.optimize.milp (uses the HiGHS solver)
- State-of-the-art commercial MILP solver: Gurobi (free academic license)

gurobi-example.py

```
import gurobipy as gp

m = gp.Model()

x = m.addVar(vtype=gp.GRB.INTEGER)
y = m.addVar(vtype=gp.GRB.INTEGER)

m.setObjective(2 * x + 3 * y, gp.GRB.MAXIMIZE)

m.addConstr(2 * x + 5 * y <= 20)
m.addConstr(3 * x + 2 * y <= 14)

m.optimize()

print(f'{m.objVal=} for {x.x=} and {y.x=}')
```

Python shell

```
| ...verbose Gurobi default output...
  m.objVal=13.0 for x.x=2.0 and y.x=3.0
```



Installing Gurobi – free academic version

- Create a user account at gurobi.com
- Get a personal free academic license for Gurobi
  (requires you to be on the university network when requesting your personal license key or that you VPN to the university network)
- Download and install the Gurobi Optimizer
- Run `grbgetkey` *`license key`* from a terminal
  (personal license key received in step 2, can also be found on your Gurobi user portal)
- `python -m pip install gurobipy`

Gurobi®

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